

Deploying Metabase on AWS

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Metabase



Disclaimer

- **No Endorsement:** This guide is for educational purposes only and does not imply endorsement by Metabase or AWS.
- **Testing Only:** Intended for testing and development, not production environments.
- **Refer to Official Docs:** Always consult official Metabase and AWS documentation for production use.
- **Proceed at Your Own Risk:** Use this guide at your own risk and be aware of potential deployment implications.

What is Metabase?



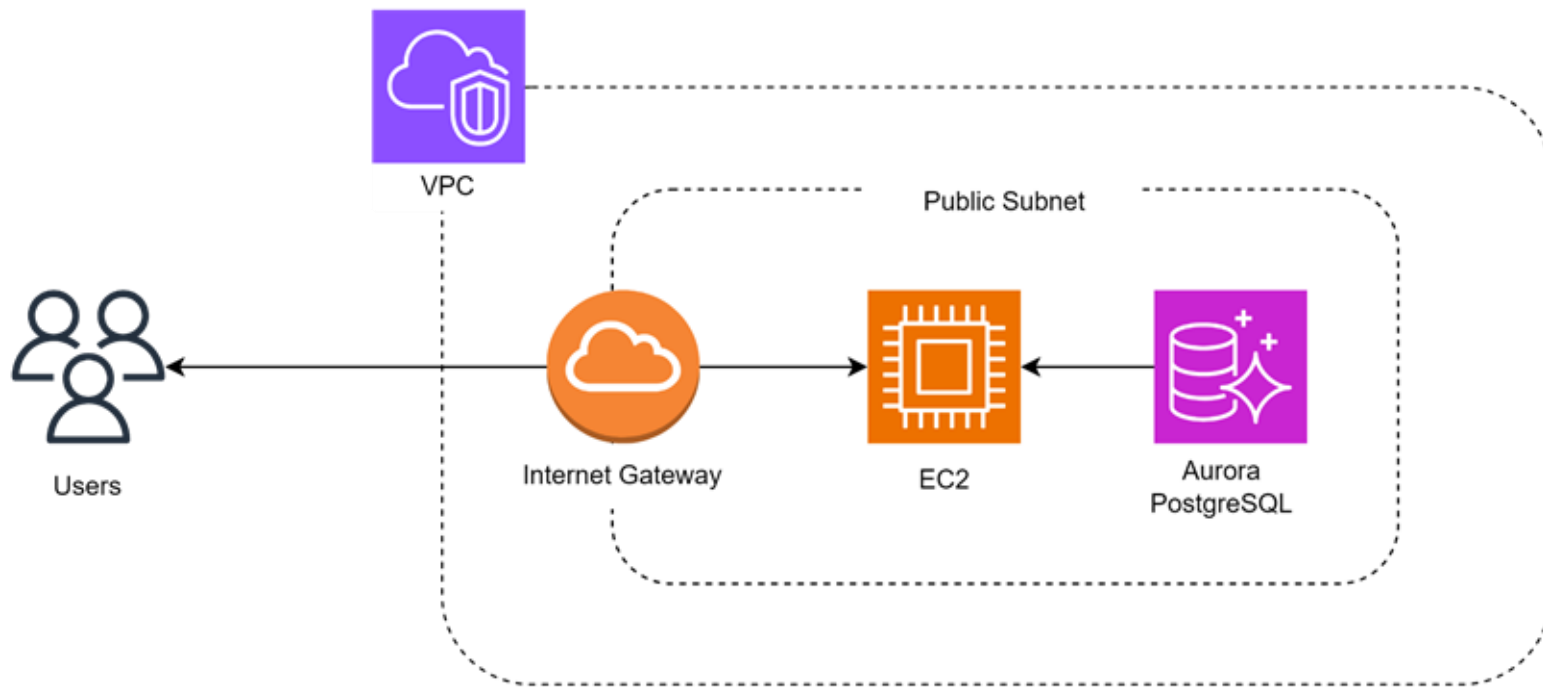
Metabase is an open-source BI tool for visualizing data through charts and dashboards. It offers a Query Builder, customizable dashboards, and alerts. Ideal for business reporting, quick analysis, and data exploration, Metabase simplifies data-driven decision-making.

Why Metabase

- **User-Friendly:** Metabase's intuitive interface lets users create and share dashboards without extensive technical knowledge.
- **Open Source:** Free and customizable, Metabase benefits from a large, active community.
- **Powerful Analytics:** Offers robust analytics, complex queries, and diverse data visualizations.
- **Seamless Integration:** Integrates smoothly with various databases and data sources.
- **Collaboration:** Promotes teamwork by enabling easy sharing of dashboards and reports.

Metabase Architecture

Metabase Architecture



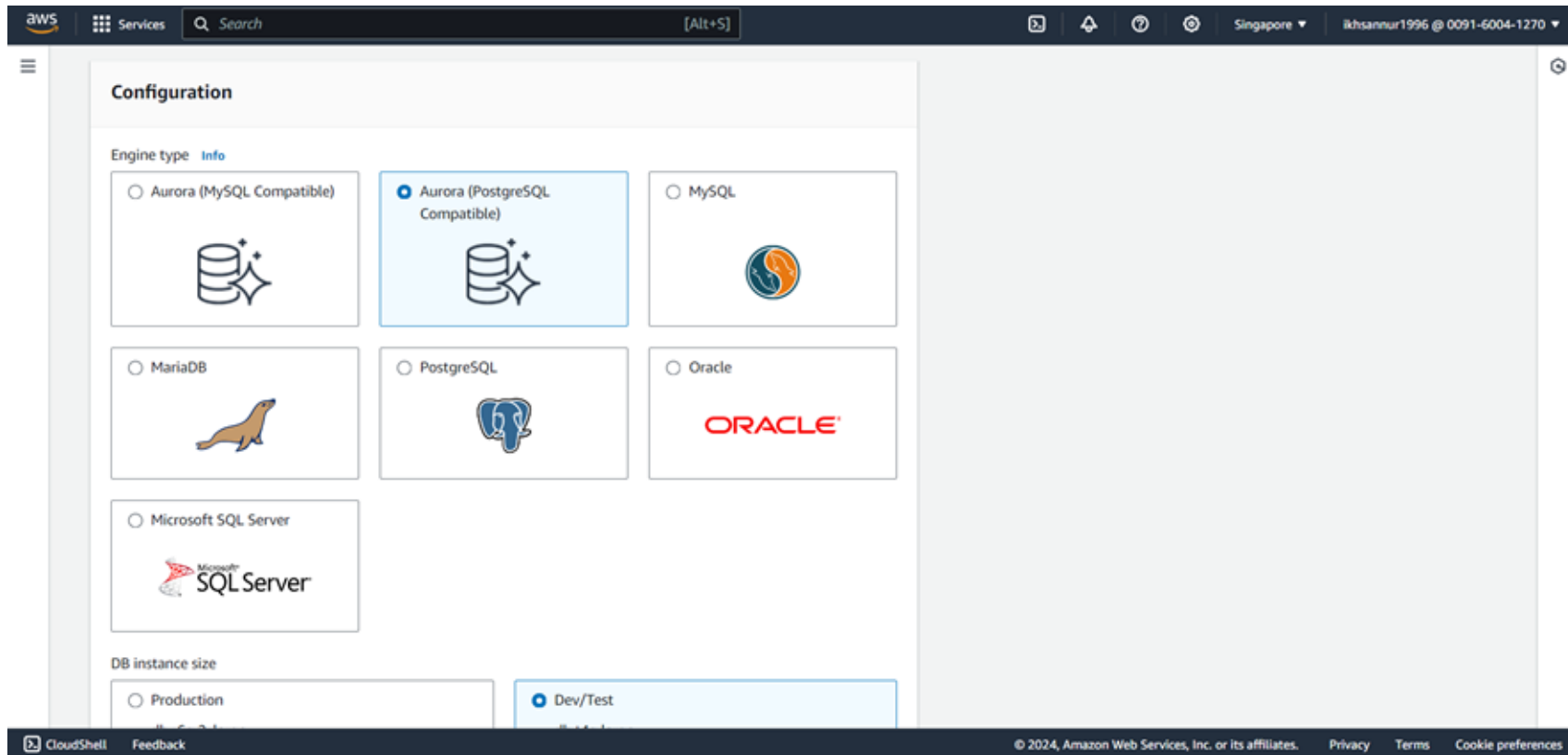
Overall Architecture: Users access the Metabase application hosted on an EC2 instance in a public subnet. The EC2 instance communicates with the Aurora PostgreSQL database within the VPC to retrieve and analyze data. The internet gateway allows the EC2 instance to be accessible from the internet, enabling users to interact with Metabase from anywhere.

Details Metabase Architecture

- **Users:** End-users (data analysts, business users) who interact with Metabase to query data, generate reports, and visualize insights.
- **VPC (Virtual Private Cloud):** A secure and scalable AWS environment for deploying resources like EC2 and RDS, ensuring network isolation and secure communication.
- **Internet Gateway:** Enables EC2 instances in the VPC to connect to the internet, allowing users to access Metabase from anywhere.
- **EC2 (Elastic Compute Cloud):** Hosts the Metabase application, processing user requests, querying databases, and serving the interface.
- **Public Subnet:** A subnet that allows the EC2 instance running Metabase to be accessible from the internet.
- **Aurora PostgreSQL:** A managed PostgreSQL database engine that stores and supports data queries and analysis in Metabase.

Setting Up RDS Aurora (PostgreSQL Compatible)

Creating RDS - Aurora (PostgreSQL Compatible)



Detailed steps for creating an Amazon RDS instance using Aurora PostgreSQL, including selecting the engine type and configuring the database instance.

Select DB Instance Size

DB instance size

☐ Production

- db.r6g.2xlarge
- 8 vCPUs
- 64 GiB RAM
- 1.253 USD/hour

☒ Dev/Test

- db.t4g.large
- 2 vCPUs
- 8 GiB RAM
- 0.229 USD/hour

DB cluster identifier

Enter a name for your DB cluster. The name must be unique across all DB clusters owned by your AWS account in the current AWS Region.

mydatabase

The DB cluster identifier is case-insensitive, but is stored as all lowercase (as in "mydbcluster"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Master username [Info](#)

Type a login ID for the master user of your DB instance.

postgres

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager to manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure

RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed

Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

password

Password strength [Info](#)

Weak

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / " ' @

Confirm master password [Info](#)

This page covers the configuration options for the DB instance, including choosing the instance size (e.g., production or dev/test), specifying the DB cluster identifier, and setting the master username.

Credentials Management

The screenshot shows the 'Create database user' page in the AWS IAM console. The page is titled 'Credentials management' and provides instructions on how to manage master user credentials. It offers two main options: 'Managed in AWS Secrets Manager - most secure' and 'Self managed'. The 'Self managed' option is selected. Below this, there is a section for 'Auto generate password' which is currently unchecked. The 'Master password' field is visible with a strength indicator showing 'Weak'. There are also fields for 'Confirm master password' and a 'Password strength' section with a 'Weak' label. At the bottom, there are links for 'Set up EC2 connection - optional' and 'View default settings for Easy create'. A disclaimer at the bottom states: 'You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.' The page includes a 'Cancel' button and a 'Create database' button.

Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength **Weak**

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

► **Set up EC2 connection - optional**
You can also set up a connection to an EC2 instance after creating the database. Go to the database list page or the database details page, choose **Actions**, and then choose **Set up to EC2 connection**.

► **View default settings for Easy create**
Easy create sets the following configurations to their default values, some of which can be changed later. If you want to change any of these settings now, use [Standard create](#).

ⓘ You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

[Cancel](#) [Create database](#)

Options for managing master user credentials, including using AWS Secrets Manager for secure password management or self-managing the credentials.

RDS Aurora PostgreSQL Success

The screenshot displays the AWS Management Console interface for an Amazon RDS instance. The left sidebar shows the navigation menu with options like Dashboard, Databases, Query Editor, and Performance insights. The main content area shows the details for the instance 'mydatabase-instance-1'.

Instance Details:

DB identifier	Status	Role	Engine	Region & AZ	Size	Recommendation
mydatabase	Available	Regional cluster	Aurora PostgreSQL	ap-southeast-1	1 instance	
mydatabase-instance-1	Available	Writer instance	Aurora PostgreSQL	ap-southeast-1c	db.t4g.large	

Connectivity & security details:

Endpoint & port	Networking	Security
Endpoint mydatabase-instance-1.cf66iaek-0a7y.ap-southeast-1.rds.amazonaws.com	Availability Zone ap-southeast-1c	VPC security groups default (sg-0aefb0b67b83dcd4)
Port 5432	VPC vpc-093bba71c48d18651	Publicly accessible No
	Subnet group default-vpc-093bba71c48d18651	Certificate authority rds-ca-rsa2048-g1

Confirmation of the successful deployment of the RDS Aurora PostgreSQL instance, with details about the database instance, such as endpoint, port, and availability zone.

Enable Public Access

The screenshot displays the AWS Management Console interface for configuring an Amazon RDS database instance. The top navigation bar includes the AWS logo, a 'Services' menu, a search bar, and the user's account information (Singapore, ikhsannur1996 @ 0091-6004-1270). The main content area is divided into two columns. The left column contains the configuration options, and the right column is currently empty.

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ **IPv4**
Your resources can communicate only over the IPv4 addressing protocol.

☐ **Dual-stack mode**
Your resources can communicate over IPv4, IPv6, or both.

DB subnet group
default-vpc-093bba71c48d18651

Security group
List of DB security groups to associate with this DB instance.
Choose security groups
default X

Certificate authority [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.
rds-ca-rsa2048-g1 (default)
Expiry: May 22, 2061

Additional configuration

Public access

☒ **Publicly accessible**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ **Not publicly accessible**
No IP address is assigned to the DB instance. EC2 instances and devices outside the VPC can't connect.

Database port
Specify the TCP/IP port that the DB instance will use for application connections. The application connection string must specify the port number. The DB security group and your firewall must allow connections to the port. [Learn more](#)
5432

The bottom of the console shows a 'CloudShell' button, a 'Feedback' link, and the copyright notice: © 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences.

Enabling public access to the database and configuring security groups.

Setting Up Amazon EC2

Launch EC2 Instance

The screenshot shows the AWS Management Console interface for launching an EC2 instance. The top navigation bar includes the AWS logo, 'Services' menu, a search bar, and user information for 'ikhsannur1996' in the 'Singapore' region. The main heading is 'Launch an instance' with an 'Info' link. Below this is a brief introduction to Amazon EC2. The 'Name and tags' section has a text input for 'Name' containing 'metabase' and a link to 'Add additional tags'. The 'Application and OS Images (Amazon Machine Image)' section includes a description of AMIs and a search bar. Below the search bar are tabs for 'Recents' and 'Quick Start', with a grid of AMI tiles for 'Amazon Linux', 'macOS', 'Ubuntu', 'Windows', 'Red Hat', and 'SUSE Li'. A 'Browse more AMIs' link is also present. On the right, the 'Summary' panel displays configuration details: 'Number of instances' set to 1, 'Software Image (AMI)' as 'Amazon Linux 2023 AMI 2023.5.2...', 'Virtual server type (instance type)' as 't2.micro', 'Firewall (security group)' as 'New security group', and 'Storage (volumes)' as '1 volume(s) - 8 GiB'. At the bottom of the summary panel are 'Cancel' and 'Launch instance' buttons, with a 'Review commands' link below the launch button. The footer contains 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
metabase [Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Li [Browse more AMIs](#)

▼ Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.5.2...[read more](#)
ami-0a6b545f62129c495

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel **Launch instance**
[Review commands](#)

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Selecting the Amazon Machine Image (AMI), instance type, and configuring security group settings.

Choose Amazon Machine Image

The screenshot shows the AWS Management Console interface for selecting an Amazon Machine Image (AMI). The top navigation bar includes the AWS logo, 'Services', a search bar, and user information. The main content area is titled 'Application and OS Images (Amazon Machine Image)' and includes a description of AMIs, a search bar, and a 'Quick Start' tab. Under 'Quick Start', there are tiles for various operating systems: Amazon Linux, macOS, Ubuntu, Windows, Red Hat, and SUSE Li. The 'Amazon Linux' tile is selected, showing details for the 'Amazon Linux 2023 AMI'. The 'Summary' section on the right provides a overview of the configuration, including the number of instances (1), the software image (AMI), the virtual server type (instance type) (t2.micro), the firewall (security group), and the storage (volumes) (1 volume(s) - 8 GiB). At the bottom of the summary section are buttons for 'Cancel', 'Launch instance', and 'Review commands'.

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents | **Quick Start**

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Li

Amazon Linux

Amazon Linux 2023 AMI
ami-0a6b545f62129c495 (64-bit (x86), uefi-preferred) / ami-080f155ef23f03e33 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.5.2...[read more](#)
ami-0a6b545f62129c495

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel **Launch instance** [Review commands](#)

Selecting the Amazon Machine Image (AMI), instance type, and configuring security group settings.

Select Instance Type

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0146 USD per Hour

On-Demand Windows base pricing: 0.0192 USD per Hour

On-Demand RHEL base pricing: 0.029 USD per Hour

On-Demand SUSE base pricing: 0.0146 USD per Hour

☒ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

Selecting the Amazon Machine Image (AMI), instance type, and configuring security group settings.

Key Pair Configuration

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended)

Default value ▼



[Create new key pair](#)

Options for managing AWS Key Pair for secure login management.

EC2 Network Configuration

Configuration of security group rules to allow access to the Metabase port, including setting up rules for HTTP, HTTPS, and SSH traffic.

▼ Network settings [Info](#)

Edit

Network [Info](#)

vpc-093bba71c48d18651

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of [free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere
0.0.0.0/0

☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

Storage Configuration

The screenshot shows the AWS Management Console interface for configuring an EC2 instance. The top navigation bar includes the AWS logo, 'Services' menu, a search bar, and the user's account information (Singapore, ikhsannur1996 @ 0091-6004-1270). A yellow warning box at the top states: 'Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.'

The main configuration area is titled 'Configure storage' with an 'Info' link and an 'Advanced' toggle. It shows a configuration for 1x 8 GiB gp3 Root volume (Not encrypted). A blue information box states: 'Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage'. Below this is an 'Add new volume' button. A section for backup information includes a refresh icon and text: 'Click refresh to view backup information. The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.' At the bottom of this section, it shows '0 x File systems' and an 'Edit' link.

The right-hand 'Summary' panel provides a quick overview: 'Number of instances' is 1; 'Software Image (AMI)' is 'Amazon Linux 2023 AMI 2023.5.2...read more' (ami-0a6b545f62129c495); 'Virtual server type (instance type)' is 't2.micro'; 'Firewall (security group)' is 'New security group'; 'Storage (volumes)' is '1 volume(s) - 8 GiB'. At the bottom of the summary, there are 'Cancel' and 'Launch instance' buttons, with a 'Review commands' link below the launch button.

The footer of the console includes 'CloudShell', 'Feedback', and copyright information: '© 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences'.

Options for Configuring Storage Size.

EC2 Instance Success Running

The screenshot displays the AWS Management Console interface for an EC2 instance. The left-hand navigation pane includes sections for EC2 Dashboard, EC2 Global View, Events, Instances, Images, Elastic Block Store, and Network & Security. The main content area is titled "Instance summary for i-03204c0ca8fe2f406 (metabase)" and shows the instance is in a "Running" state. The summary is organized into three columns: Instance ID, Public IPv4 address, Private IPv4 addresses, Instance state, Hostname type, Private IP DNS name, Instance type, VPC ID, Subnet ID, Instance ARN, Answer private resource DNS name, Auto-assigned IP address, IAM Role, IMDSv2, and Elastic IP addresses. The instance is a t2.micro type in the vpc-093bba71c48d18651 VPC, using subnet-0fabf15788bfd73f. The instance state is "Running" with a green checkmark icon. The public IP address is 175.41.178.121, and the private IP address is 172.31.37.128. The instance is associated with the IAM Role "Required" and the IMDSv2 setting is "Required". The instance is also associated with the Elastic IP address "ec2-175-41-178-121.ap-southeast-1.compute.amazonaws.com".

Instance ID	Public IPv4 address	Private IPv4 addresses
i-03204c0ca8fe2f406 (metabase)	175.41.178.121 open address	172.31.37.128
IPv6 address	Instance state	Public IPv4 DNS
-	Running	ec2-175-41-178-121.ap-southeast-1.compute.amazonaws.com open address
Hostname type	Private IP DNS name (IPv4 only)	Elastic IP addresses
IP name: ip-172-31-37-128.ap-southeast-1.compute.internal	ip-172-31-37-128.ap-southeast-1.compute.internal	-
Answer private resource DNS name	Instance type	AWS Compute Optimizer finding
IPv4 (A)	t2.micro	Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address	VPC ID	Auto Scaling Group name
175.41.178.121 (Public IP)	vpc-093bba71c48d18651	-
IAM Role	Subnet ID	
-	subnet-0fabf15788bfd73f	
IMDSv2	Instance ARN	
Required	arn:aws:ec2:ap-southeast-1:009160041270:instance/i-03204c0ca8fe2f406	

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Confirmation of the successful launch of the EC2 instance, with details about the instance, such as instance ID, public and private IP addresses, and instance state.

Custom Inbound Rules for Metabase Port

The screenshot displays the AWS Management Console interface for a security group named 'sg-053a5c2740783d442 - launch-wizard-1'. The left sidebar shows the navigation menu with categories like EC2 Dashboard, Instances, Images, Elastic Block Store, and Network & Security. The main content area is titled 'sg-053a5c2740783d442 - launch-wizard-1' and includes an 'Actions' button. Below the title, the 'Details' section provides information about the security group, including its name, ID, description, VPC ID, owner, and rule counts. The 'Inbound rules' tab is selected, showing a list of four inbound rules. Each rule is configured to allow traffic from all sources (0.0.0.0/0) on specific ports: 80 (HTTP), 3000 (Custom TCP), 443 (HTTPS), and 22 (SSH). The rules are listed in a table with columns for Name, Security group rule ID, IP version, Type, Protocol, and Port range.

Details

Security group name launch-wizard-1	Security group ID sg-053a5c2740783d442	Description launch-wizard-1 created 2024-08-14T12:00:18.750Z	VPC ID vpc-093bba71c48d18651
Owner 009160041270	Inbound rules count 4 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules (4)

☐	Name	Security group rule...	IP version	Type	Protocol	Port range
☐	-	sgr-0aff923425986d2a2	IPv4	HTTP	TCP	80
☐	-	sgr-0ea0f8bbb34ea16d2	IPv4	Custom TCP	TCP	3000
☐	-	sgr-03432b26f12a7bd...	IPv4	HTTPS	TCP	443
☐	-	sgr-01b99c25242a461f7	IPv4	SSH	TCP	22

Configuration of security group rules to allow access to the Metabase port, including setting up rules for HTTP, HTTPS, and SSH traffic.

Connect to Instance

The screenshot shows the AWS 'Connect to instance' page. At the top, there's a navigation bar with the AWS logo, 'Services', a search bar, and a keyboard shortcut '[Alt+5]'. On the right, it shows the location 'Singapore' and the user 'ikhsannur1996 @ 0091-6004-1270'. Below the navigation bar, the main heading is 'Connect to instance' with an 'Info' link. A subtitle reads: 'Connect to your instance i-03204c0ca8fe2f406 (metabase) using any of these options'. There are four tabs: 'EC2 Instance Connect' (selected), 'Session Manager', 'SSH client', and 'EC2 serial console'. A yellow warning box states: 'Port 22 (SSH) is open to all IPv4 addresses. Port 22 (SSH) is currently open to all IPv4 addresses, indicated by 0.0.0.0/0 in the inbound rule in your security group. For increased security, consider restricting access to only the EC2 Instance Connect service IP addresses for your Region: 3.0.5.32/29. Learn more.' Below this, the instance details are shown: 'Instance ID' is 'i-03204c0ca8fe2f406 (metabase)'. Under 'Connection Type', there are two options: 'Connect using EC2 Instance Connect' (selected, described as 'Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.') and 'Connect using EC2 Instance Connect Endpoint' (described as 'Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.'). The 'Public IP address' is '175.41.178.121'. The 'Username' field is 'ec2-user', with a note: 'Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.' A blue note box at the bottom says: 'Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.' At the bottom right are 'Cancel' and 'Connect' buttons. The footer contains 'CloudShell', 'Feedback', and copyright information: '© 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences'.

Connect to instance [Info](#)

Connect to your instance i-03204c0ca8fe2f406 (metabase) using any of these options

[EC2 Instance Connect](#) Session Manager SSH client EC2 serial console

Warning: Port 22 (SSH) is open to all IPv4 addresses
Port 22 (SSH) is currently open to all IPv4 addresses, indicated by 0.0.0.0/0 in the inbound rule in [your security group](#). For increased security, consider restricting access to only the EC2 Instance Connect service IP addresses for your Region: 3.0.5.32/29. [Learn more](#).

Instance ID
i-03204c0ca8fe2f406 (metabase)

Connection Type

☒ **Connect using EC2 Instance Connect**
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.

☐ **Connect using EC2 Instance Connect Endpoint**
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IP address
175.41.178.121

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.
ec2-user

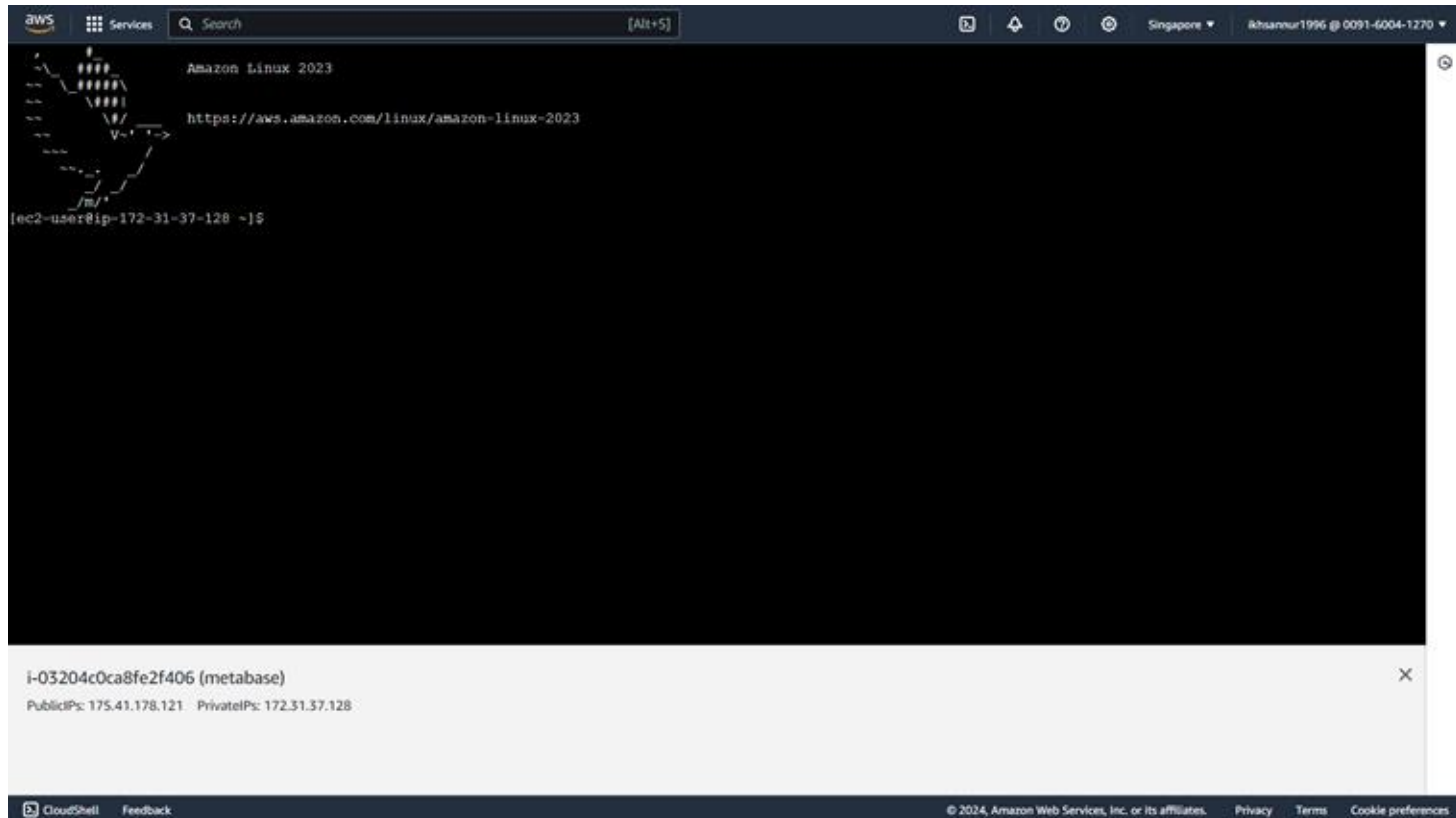
Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel **Connect**

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Instructions for connecting to the EC2 instance using various methods, such as EC2 Instance Connect, Session Manager, SSH client, and EC2 serial console.

SSH Opened



Setting Up Metabase on Amazon EC2

METABASE OPEN SOURCE

Run your own free self-hosted instance of Metabase

Docker Image

Metabase provides an official Docker image via Dockerhub that can be used for deployments on any system that is running Docker. Here's a one-liner that will start a container running Metabase.

```
docker run -d -p 3000:3000 --name metabase metabase/metabase
```

[Running on Docker documentation >](#)

JAR file

1. Download Metabase.
2. Create a new directory and move the Metabase JAR into it.
3. Change into your new Metabase directory and run the JAR.

```
java -jar metabase.jar
```

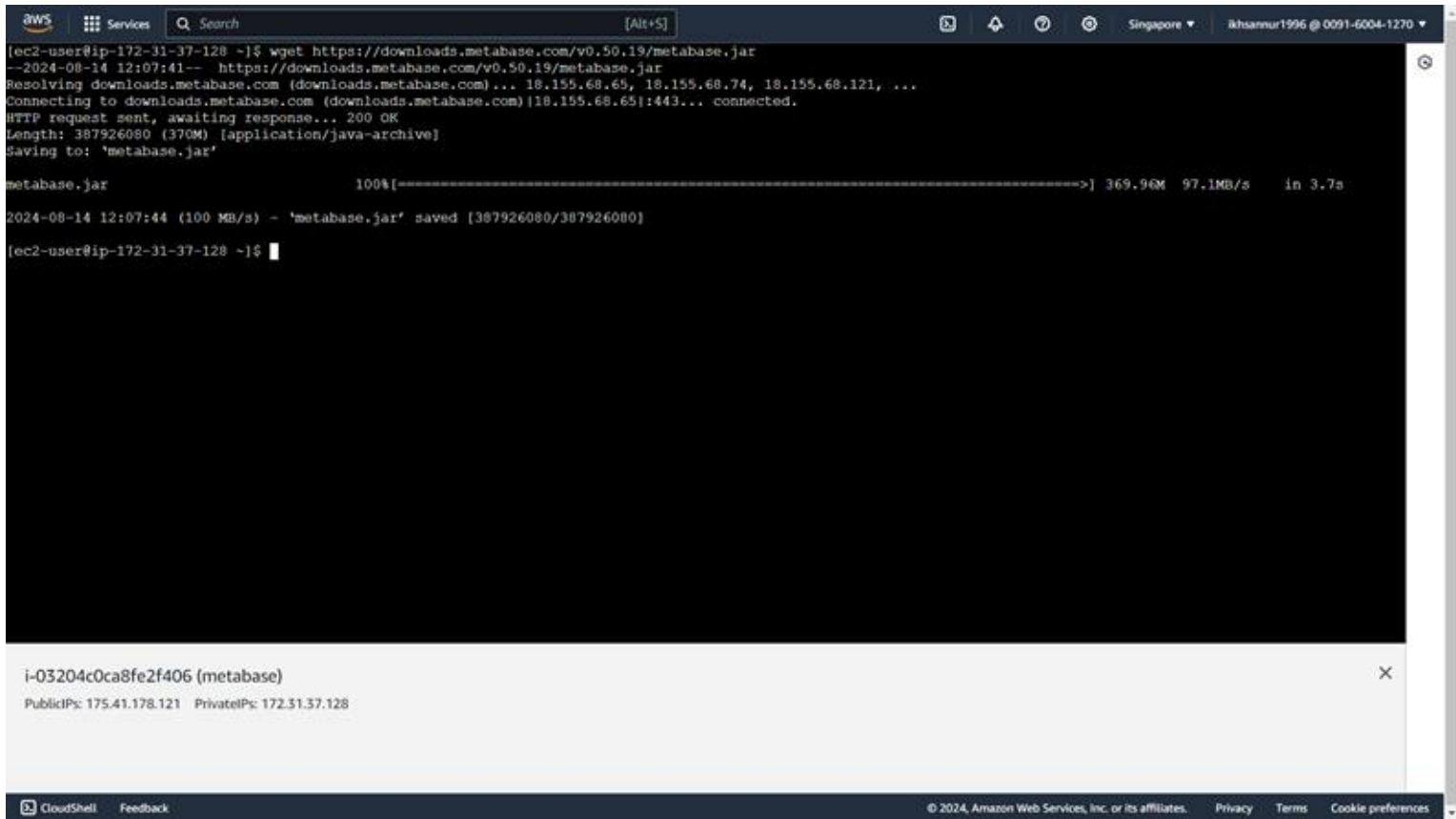
Metabase will log its progress in the terminal as it starts up. Wait until you see "Metabase Initialization Complete" and visit localhost:3000.

[Running the Jar File documentation >](#)

Steps to install Metabase on the EC2 instance, including downloading the Metabase application and setting up the necessary configurations. If Java isn't installed, you'll need to install Java before you can run Metabase.

<https://www.metabase.com/start/oss/>

Downloading Metabase



The screenshot shows an AWS CloudShell terminal window. The top bar includes the AWS logo, 'Services', a search bar, and the user's location 'Singapore' and account ID 'ikhsanur1996 @ 0091-6004-1270'. The terminal output shows the execution of the 'wget' command to download 'metabase.jar' from 'https://downloads.metabase.com/v0.50.19/metabase.jar'. The download progress is shown as 100% completed, with a size of 369.96M and a download speed of 97.1MB/s. The file is saved to the local directory. Below the terminal output, a summary box for the instance 'i-03204c0ca8fe2f406 (metabase)' is displayed, showing public and private IP addresses. The bottom bar contains the 'CloudShell' logo, a 'Feedback' link, and copyright information for Amazon Web Services.

```
aws
Services
[Alt+S]
Singapore
ikhsanur1996 @ 0091-6004-1270

[ec2-user@ip-172-31-37-128 ~]$ wget https://downloads.metabase.com/v0.50.19/metabase.jar
--2024-08-14 12:07:41-- https://downloads.metabase.com/v0.50.19/metabase.jar
Resolving downloads.metabase.com (downloads.metabase.com)... 18.155.68.65, 18.155.68.74, 18.155.68.121, ...
Connecting to downloads.metabase.com (downloads.metabase.com)|18.155.68.65|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 387926080 (370M) [application/java-archive]
Saving to: 'metabase.jar'

metabase.jar                               100%[=====>] 369.96M  97.1MB/s   in 3.7s

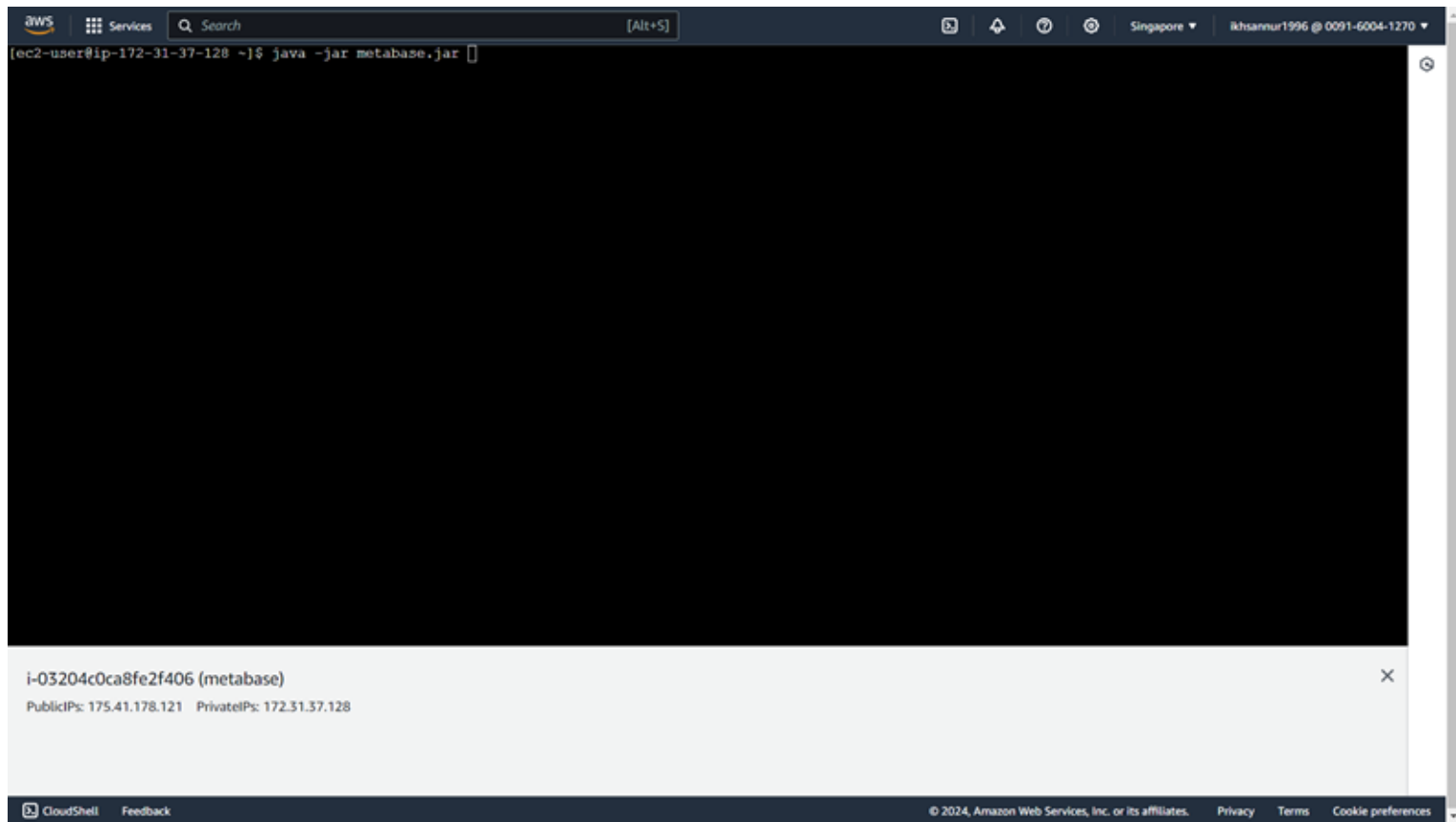
2024-08-14 12:07:44 (100 MB/s) - 'metabase.jar' saved [387926080/387926080]

[ec2-user@ip-172-31-37-128 ~]$
```

i-03204c0ca8fe2f406 (metabase)
PublicIPs: 175.41.178.121 PrivateIPs: 172.31.37.128

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Run The Metabase



Metabase Success Running

```
aws Services [Alt+S] Singapore | kshannur1996 @ 0091-6004-1270
2024-08-14 12:12:53,724 INFO jdbcjobstore.JobStoreTX :: JobStoreTX initialized.
2024-08-14 12:12:53,724 INFO core.QuartzScheduler :: Scheduler metabase-data: Quartz Scheduler (v2.3.2) 'MetabaseScheduler' with instanceId 'ip-172-31-37-128.ap-south-east-1.compute.internal1723637573699'
Scheduler class: 'org.quartz.core.QuartzScheduler' - running locally.
NOT STARTED.
Currently in standby mode.
Number of jobs executed: 0
Using thread pool 'org.quartz.simpl.SimpleThreadPool' - with 10 threads.
Using job-store 'org.quartz.impl.jdbcjobstore.JobStoreTX' - which supports persistence, and is clustered.

2024-08-14 12:12:53,725 INFO impl.StdSchedulerFactory :: Quartz scheduler 'MetabaseScheduler' initialized from default resource file in Quartz package: 'quartz.properties'
2024-08-14 12:12:53,725 INFO impl.StdSchedulerFactory :: Quartz scheduler version: 2.3.2
2024-08-14 12:12:53,914 INFO core.QuartzScheduler :: Scheduler MetabaseScheduler_5_ip-172-31-37-128.ap-southeast-1.compute.internal1723637573699 paused.
2024-08-14 12:12:53,916 INFO metabase.task :: Task scheduler initialized into standby mode.
2024-08-14 12:12:53,920 INFO metabase.task :: Initializing task Cache
2024-08-14 12:12:53,923 INFO metabase.task :: Initializing task SyncDatabases
2024-08-14 12:12:53,969 INFO task.sync-databases :: Updated default schedules for 0 databases
2024-08-14 12:12:53,969 INFO metabase.task :: Initializing task PersistTheFresh
2024-08-14 12:12:54,018 INFO metabase.task :: Initializing task CheckForNewVersions
2024-08-14 12:12:54,031 INFO metabase.task :: Initializing task PersistFrupe
2024-08-14 12:12:54,032 INFO metabase.task :: Initializing task SendAnonymousUsageStats
2024-08-14 12:12:54,038 INFO metabase.task :: Initializing task ModelIndexValues
2024-08-14 12:12:54,039 INFO metabase.task :: Initializing task RefreshSlackChannelsAndUsers
2024-08-14 12:12:54,051 INFO metabase.task :: Initializing task TruncateAuditTables
2024-08-14 12:12:54,054 INFO metabase.task :: Initializing task SendPulse
2024-08-14 12:12:54,064 INFO metabase.task :: Initializing task SendFollowUpEmails
2024-08-14 12:12:54,066 INFO metabase.task :: Initializing task SendCreatorSentimentEmails
2024-08-14 12:12:54,080 INFO metabase.task :: Initializing task SendLegacyNoSelfServiceEmail
2024-08-14 12:12:54,081 INFO metabase.task :: Initializing task TaskHistoryCleanup
2024-08-14 12:12:54,083 INFO metabase.task :: Initializing task SendWarnPulseRemovalEmail
2024-08-14 12:12:54,094 INFO core.QuartzScheduler :: Scheduler MetabaseScheduler_5_ip-172-31-37-128.ap-southeast-1.compute.internal1723637573699 started.
2024-08-14 12:12:54,101 INFO metabase.task :: Task scheduler started
2024-08-14 12:12:54,101 INFO metabase.core :: Metabase Initialization COMPLETE in 1.0 mins
2024-08-14 12:12:54,137 INFO task.refresh-slack-channel-user-cache :: Slack is not configured, not refreshing slack user/channel cache.

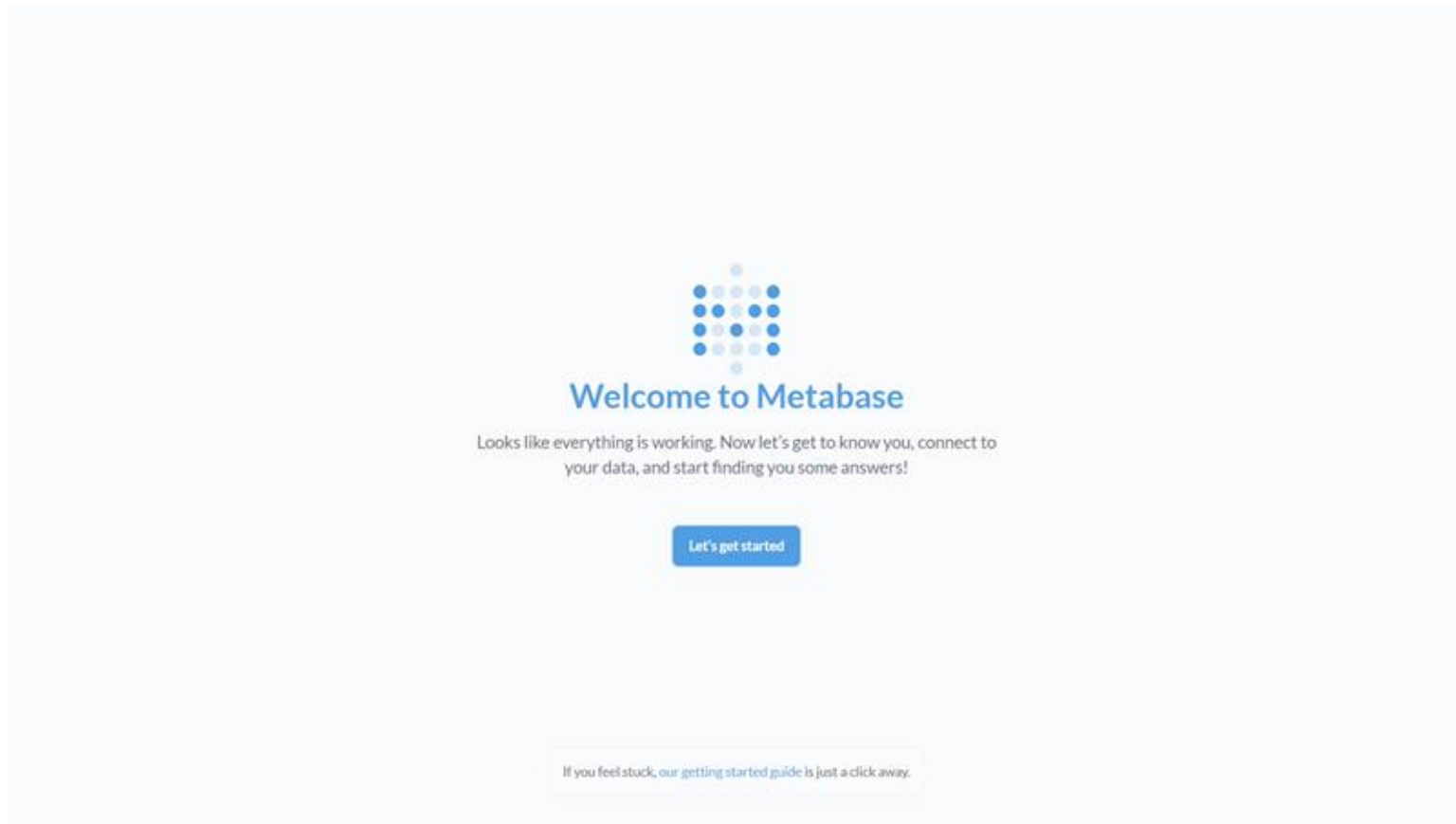
i-03204c0ca8fe2f406 (metabase)
PublicIPs: 175.41.178.121 PrivateIPs: 172.31.37.128

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```

Logs and details of the Metabase initialization process, including the setup of the Quartz scheduler and various Metabase tasks. Metabase will run in port 3000.

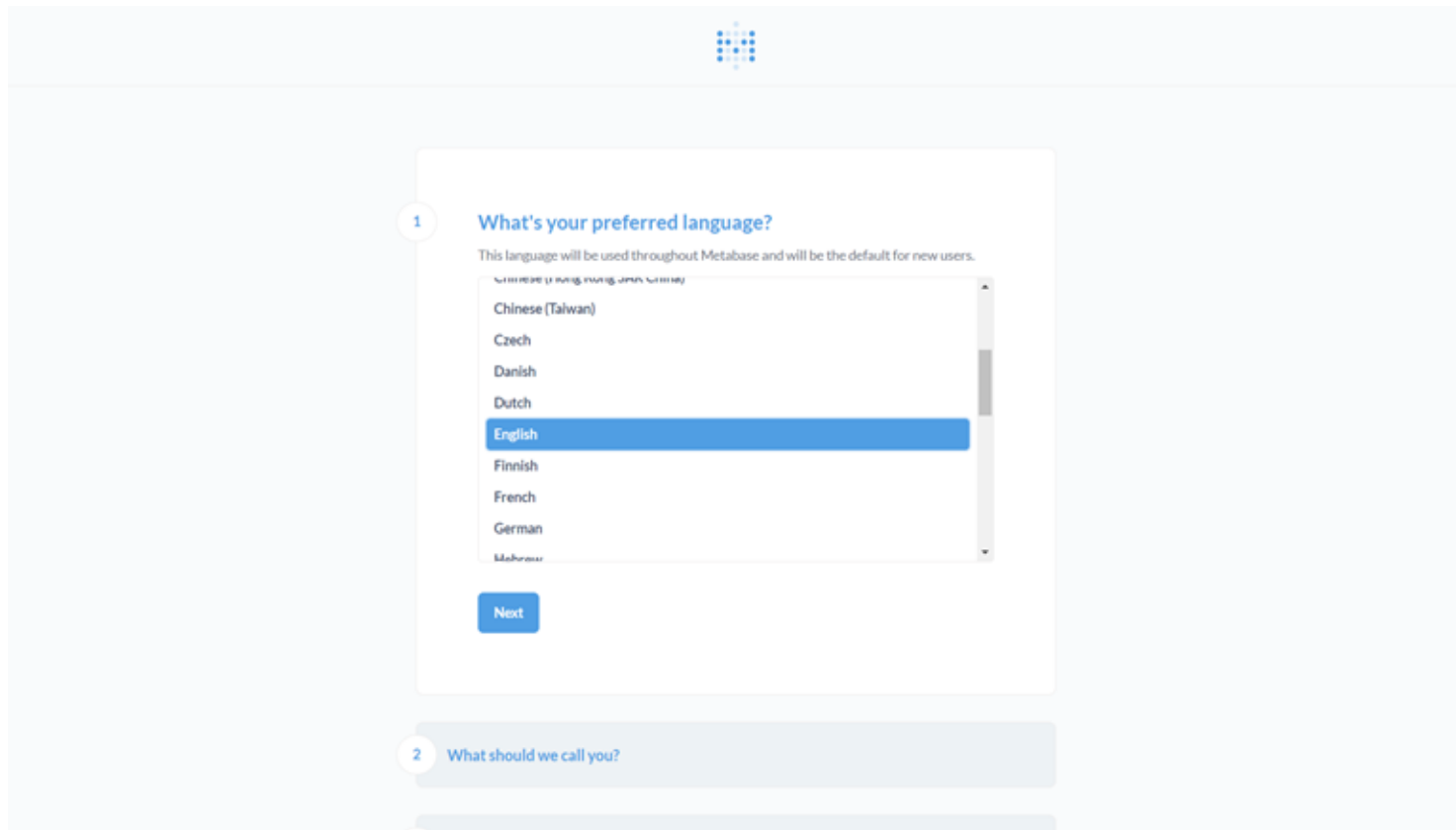
Metabase Initialisation and Configuration

Welcome to Metabase



Initial setup steps for Metabase, including selecting the preferred language, entering user information, and setting up a password.

Language Configuration



The image shows a screenshot of the Metabase initial setup interface. At the top center is the Metabase logo, a 3x3 grid of blue squares. Below it is a white card with a light blue border. The card has a small circle with the number '1' on its left side. The title of the card is 'What's your preferred language?'. Below the title is a subtitle: 'This language will be used throughout Metabase and will be the default for new users.' Below the subtitle is a scrollable list of languages. The languages listed are: Chinese (Taiwan), Czech, Danish, Dutch, English (highlighted with a blue background), Finnish, French, German, and Hebrew. Below the list is a blue button labeled 'Next'. Below the card is a light blue bar with a small circle containing the number '2' and the text 'What should we call you?'. Below this bar is another light blue bar.

1 What's your preferred language?

This language will be used throughout Metabase and will be the default for new users.

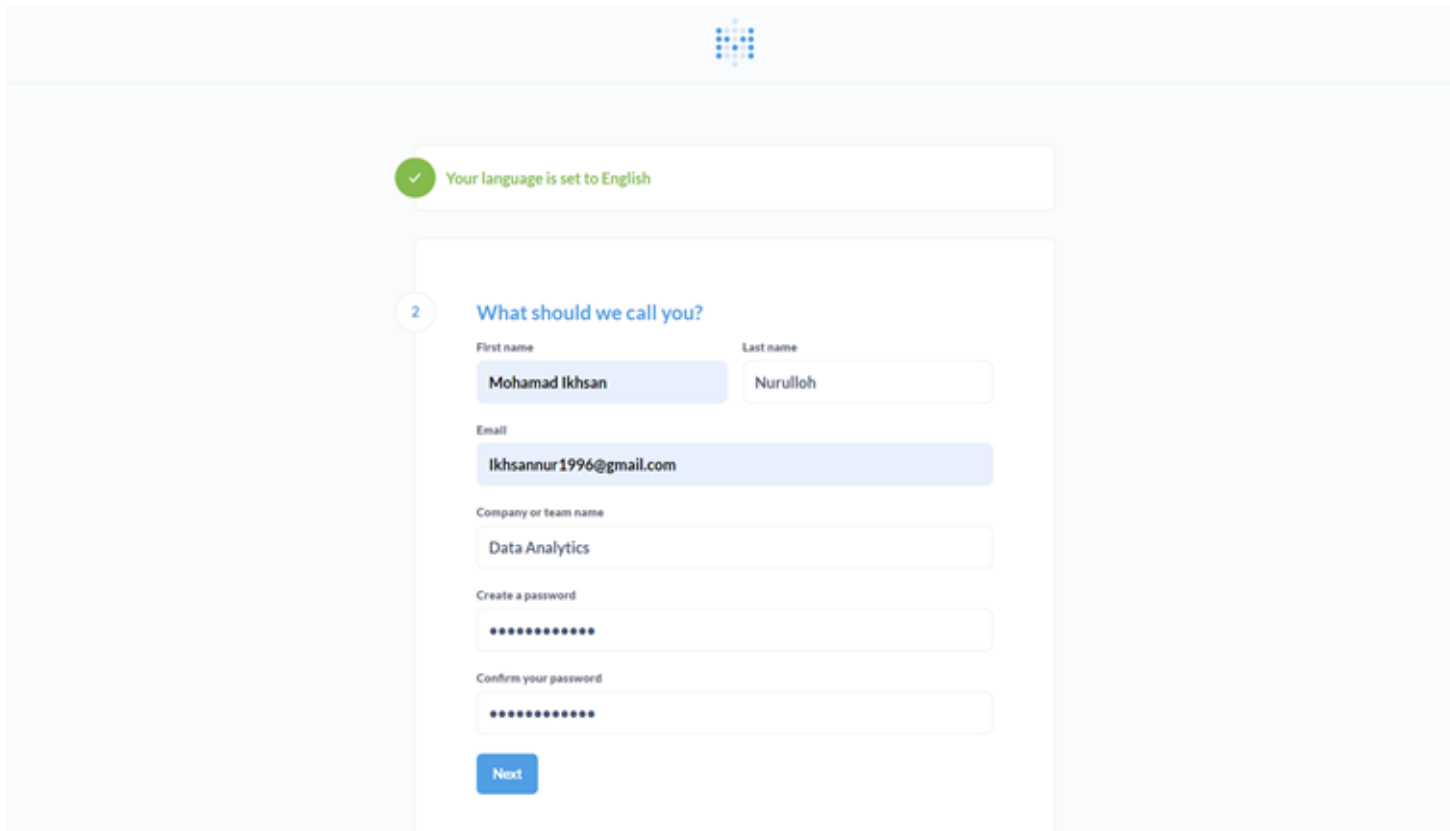
Chinese (Taiwan)
Czech
Danish
Dutch
English
Finnish
French
German
Hebrew

Next

2 What should we call you?

Initial setup steps for Metabase, including selecting the preferred language, entering user information, and setting up a password.

User Information and Setting Up a Password



The image shows the initial setup steps for Metabase. At the top, there is a Metabase logo. Below it, a green checkmark icon and the text "Your language is set to English" are displayed. The main form is titled "2 What should we call you?". It contains several input fields: "First name" with the value "Mohamad Ikhsan", "Last name" with the value "Nurulloh", "Email" with the value "lkhsannur1996@gmail.com", "Company or team name" with the value "Data Analytics", "Create a password" with a masked password "*****", and "Confirm your password" with a masked password "*****". A blue "Next" button is located at the bottom of the form.

✓ Your language is set to English

2 What should we call you?

First name Last name

Mohamad Ikhsan Nurulloh

Email

lkhsannur1996@gmail.com

Company or team name

Data Analytics

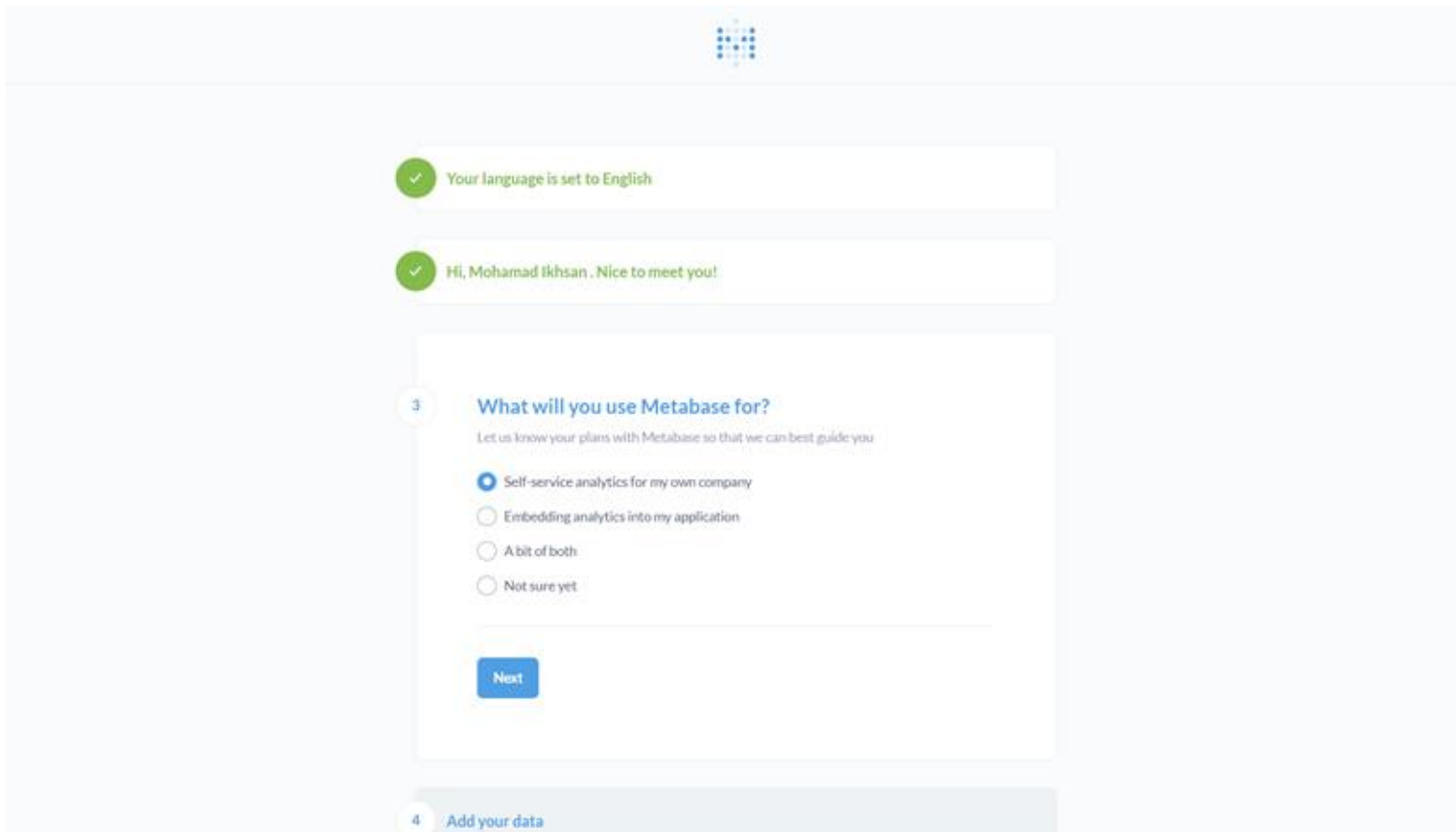
Create a password

Confirm your password

Next

Initial setup steps for Metabase, including selecting the preferred language, entering user information, and setting up a password.

What will you use Metabase for?



The image shows a Metabase onboarding interface. At the top center is the Metabase logo, a 4x4 grid of blue dots. Below it are two green checkmark notifications: 'Your language is set to English' and 'Hi, Mohamad Ikhsan, Nice to meet you!'. The main content area is a white card with a blue border. It has a step indicator '3' in a circle on the left. The title is 'What will you use Metabase for?' in blue. Below the title is a subtitle: 'Let us know your plans with Metabase so that we can best guide you'. There are four radio button options: 'Self-service analytics for my own company' (selected), 'Embedding analytics into my application', 'A bit of both', and 'Not sure yet'. At the bottom of the card is a blue 'Next' button. At the very bottom of the screen is a light blue bar with a step indicator '4' in a circle and the text 'Add your data'.

3 What will you use Metabase for?

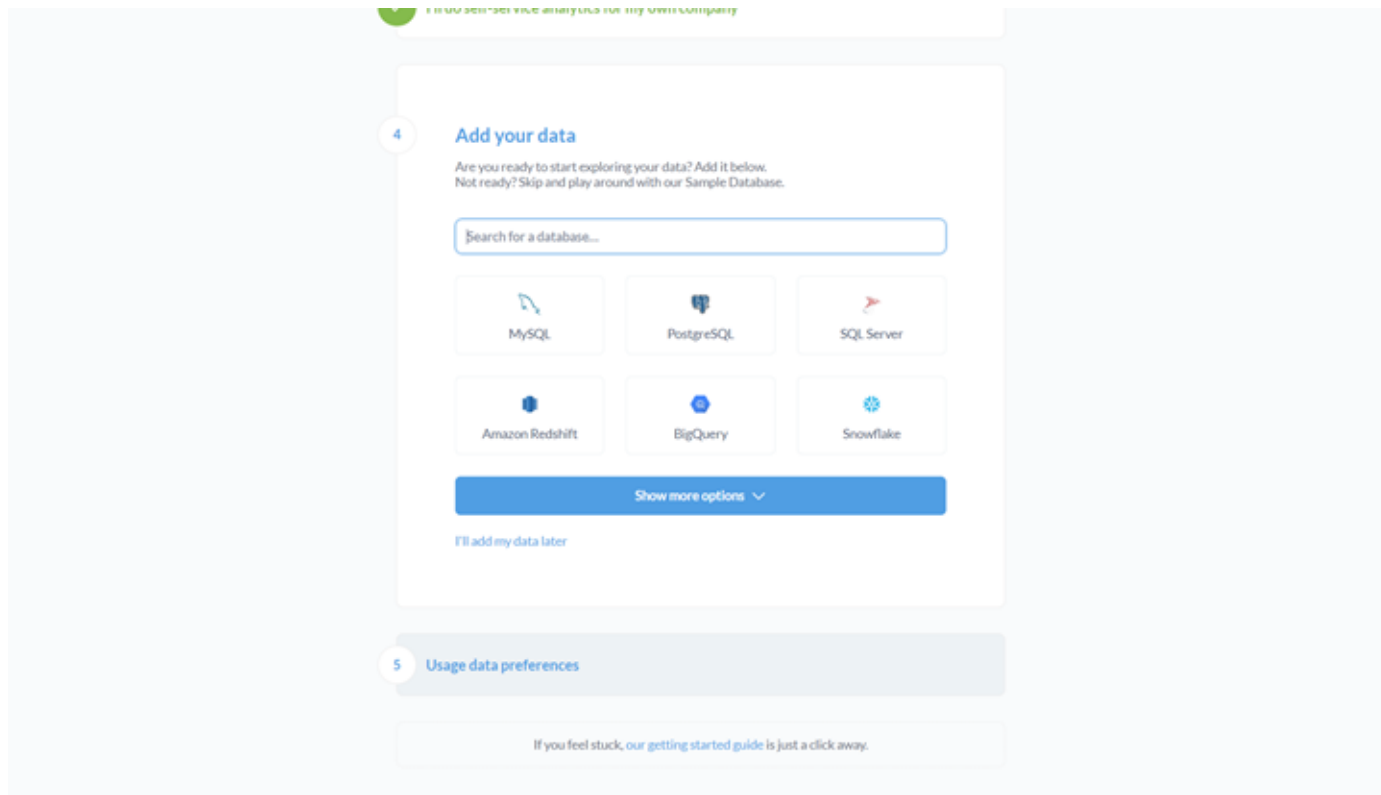
Let us know your plans with Metabase so that we can best guide you

- ☒ Self-service analytics for my own company
- ☐ Embedding analytics into my application
- ☐ A bit of both
- ☐ Not sure yet

Next

4 Add your data

Add Data Connection



Options for adding data sources to Metabase, including popular databases like MySQL, PostgreSQL, SQL Server, Amazon Redshift, BigQuery, and Snowflake.

Usage Data Preferences



Your language is set to English



Hi, Mohamad Ikhsan. Nice to meet you!



I'll do self-service analytics for my own company



I'll add my own data later

5

Usage data preferences

In order to help us improve Metabase, we'd like to collect certain data about product usage. [Here's a full list of what we track and why.](#)



Allow Metabase to anonymously collect usage events

Finish

If you feel stuck, our [getting started guide](#) is just a click away.

You're All Set Up



I'll do self-service analytics for my own company



I'll add my own data later



We won't collect any usage events

You're all set up!

 METABASE NEWSLETTER

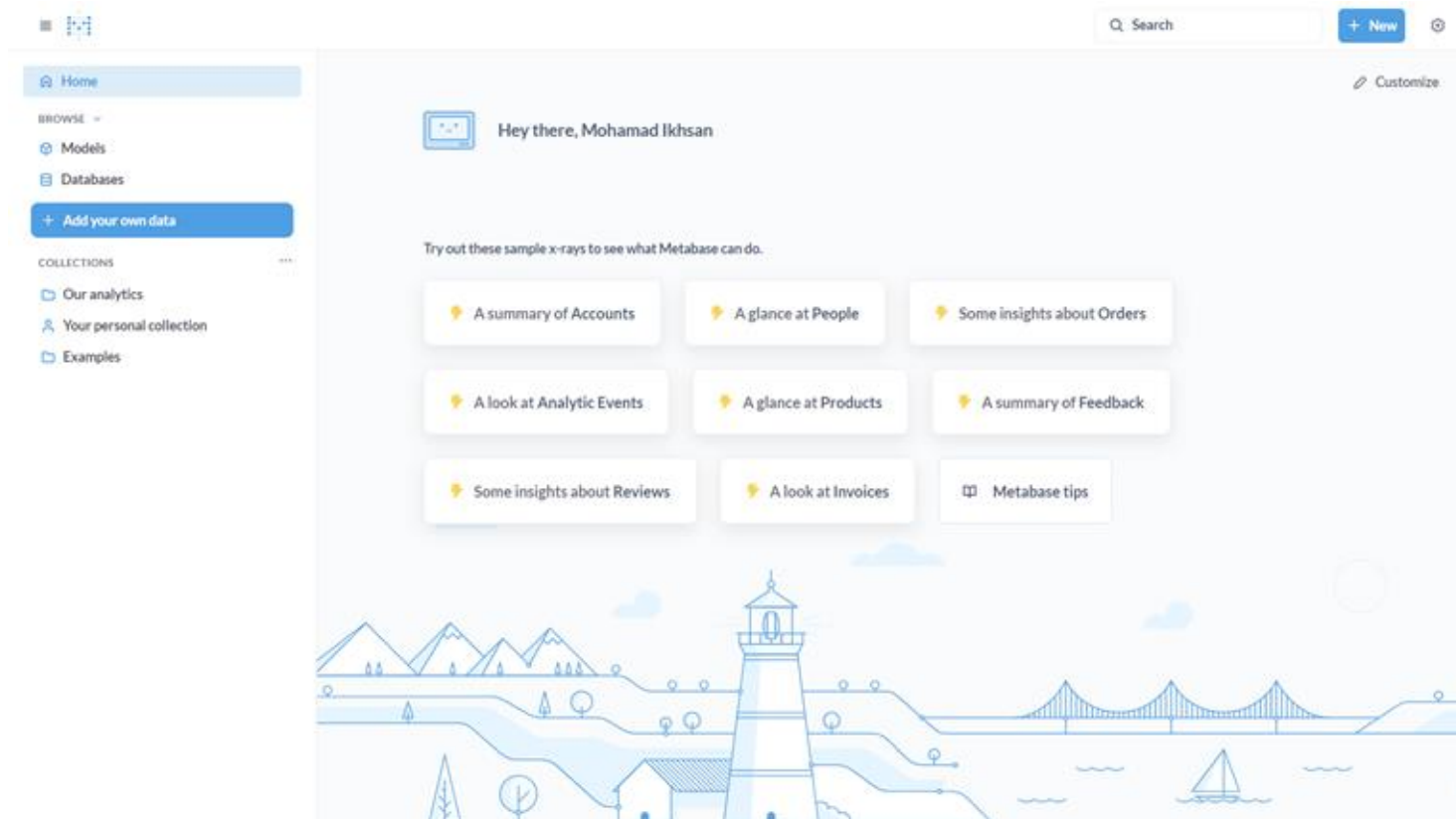
Get infrequent emails about new releases and feature updates.

Subscribe

Take me to Metabase

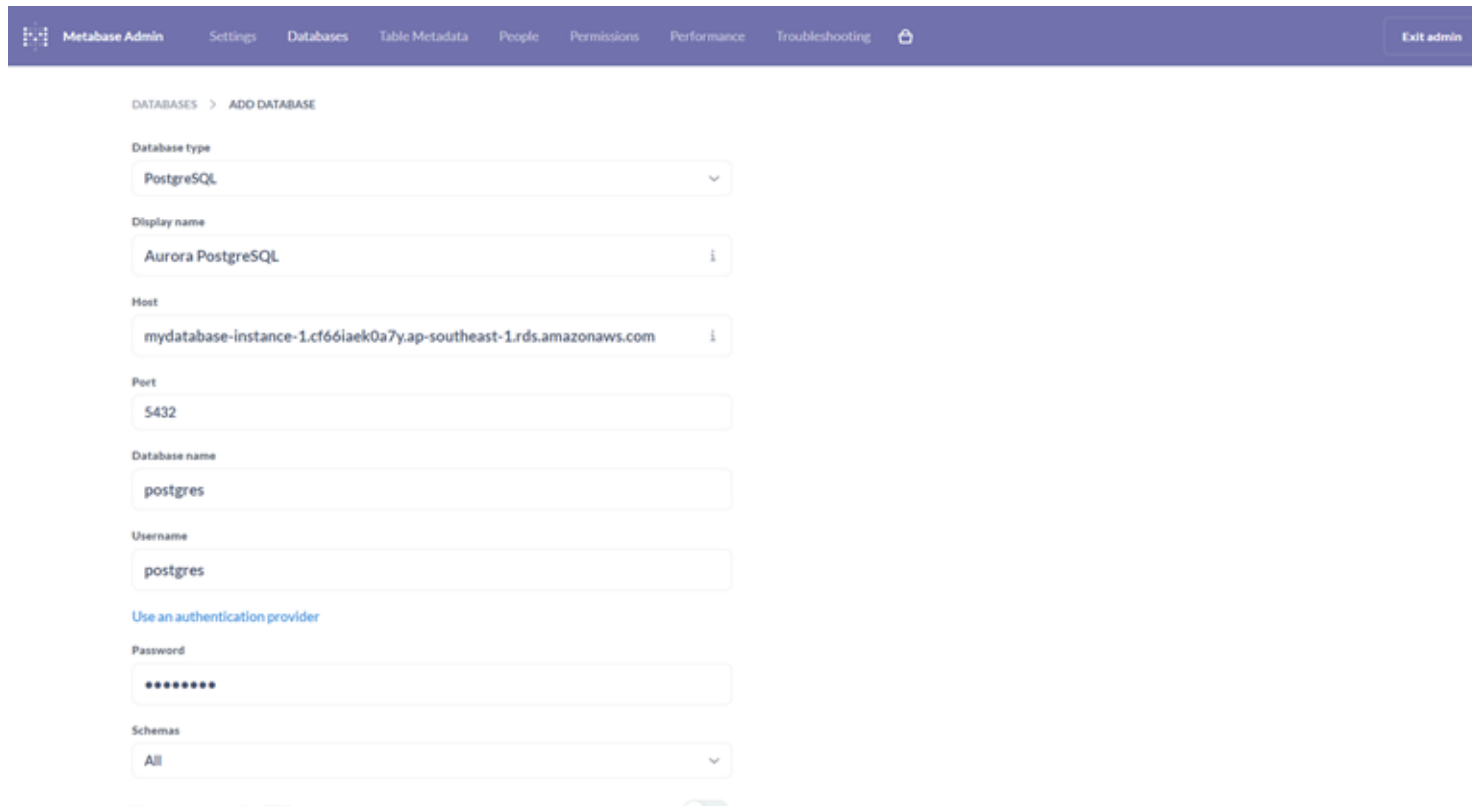
If you feel stuck, [our getting started guide](#) is just a click away.

Metabase Home Page



Overview of the Metabase dashboard, including sample data exploration options and a summary of various analytics insights.

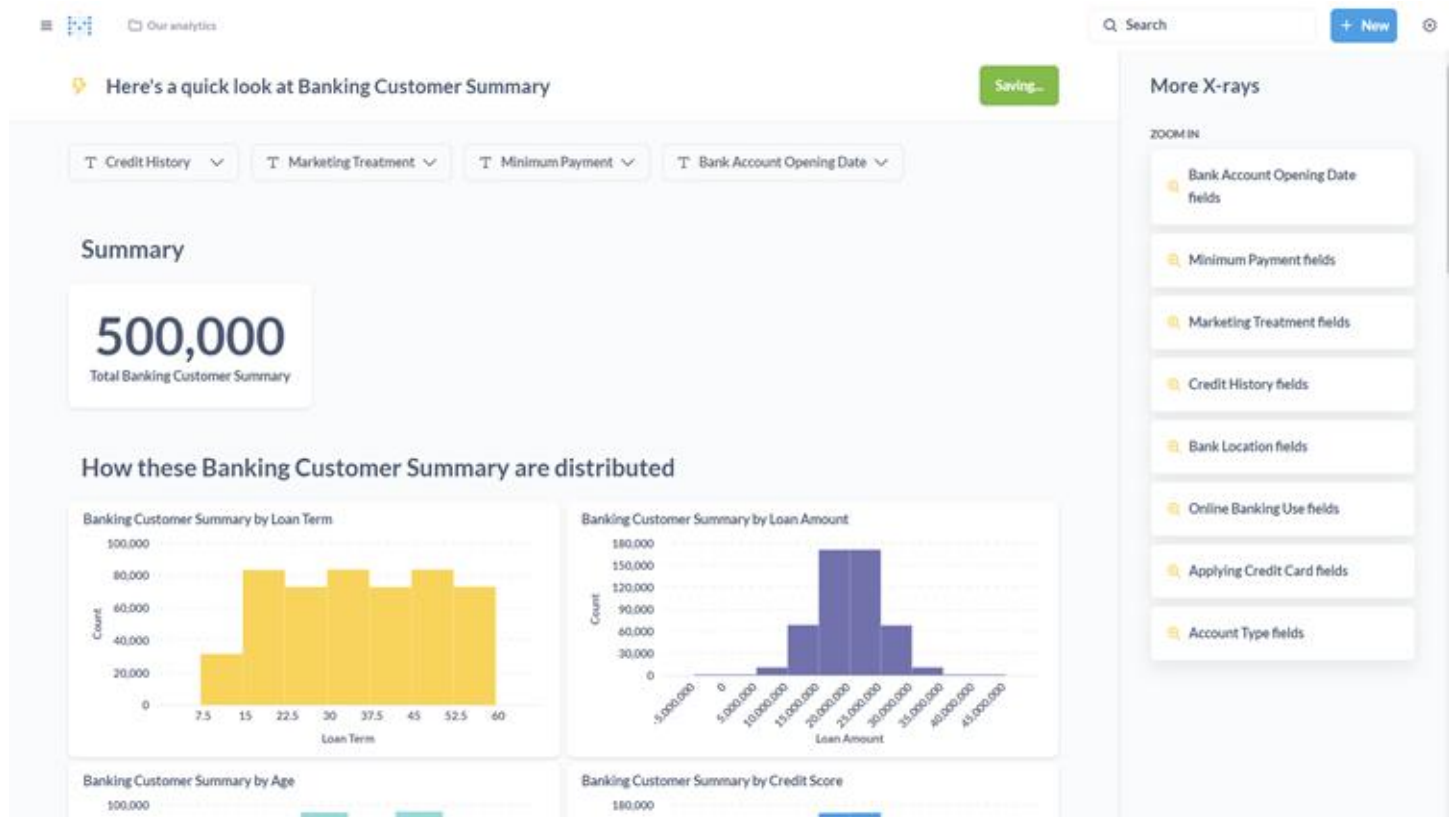
Test Database Connection to Aurora PostgreSQL



The screenshot shows the Metabase Admin interface with a purple header bar. The header contains navigation links: Metabase Admin, Settings, Databases, Table Metadata, People, Permissions, Performance, Troubleshooting, and an Exit admin button. The main content area is titled 'DATABASES > ADD DATABASE'. It contains a form with the following fields: 'Database type' (a dropdown menu set to 'PostgreSQL'), 'Display name' (a text input field containing 'Aurora PostgreSQL'), 'Host' (a text input field containing 'mydatabase-instance-1.cf66laek0a7y.ap-southeast-1.rds.amazonaws.com'), 'Port' (a text input field containing '5432'), 'Database name' (a text input field containing 'postgres'), 'Username' (a text input field containing 'postgres'), 'Use an authentication provider' (a blue link), 'Password' (a password input field with masked characters), and 'Schemas' (a dropdown menu set to 'All').

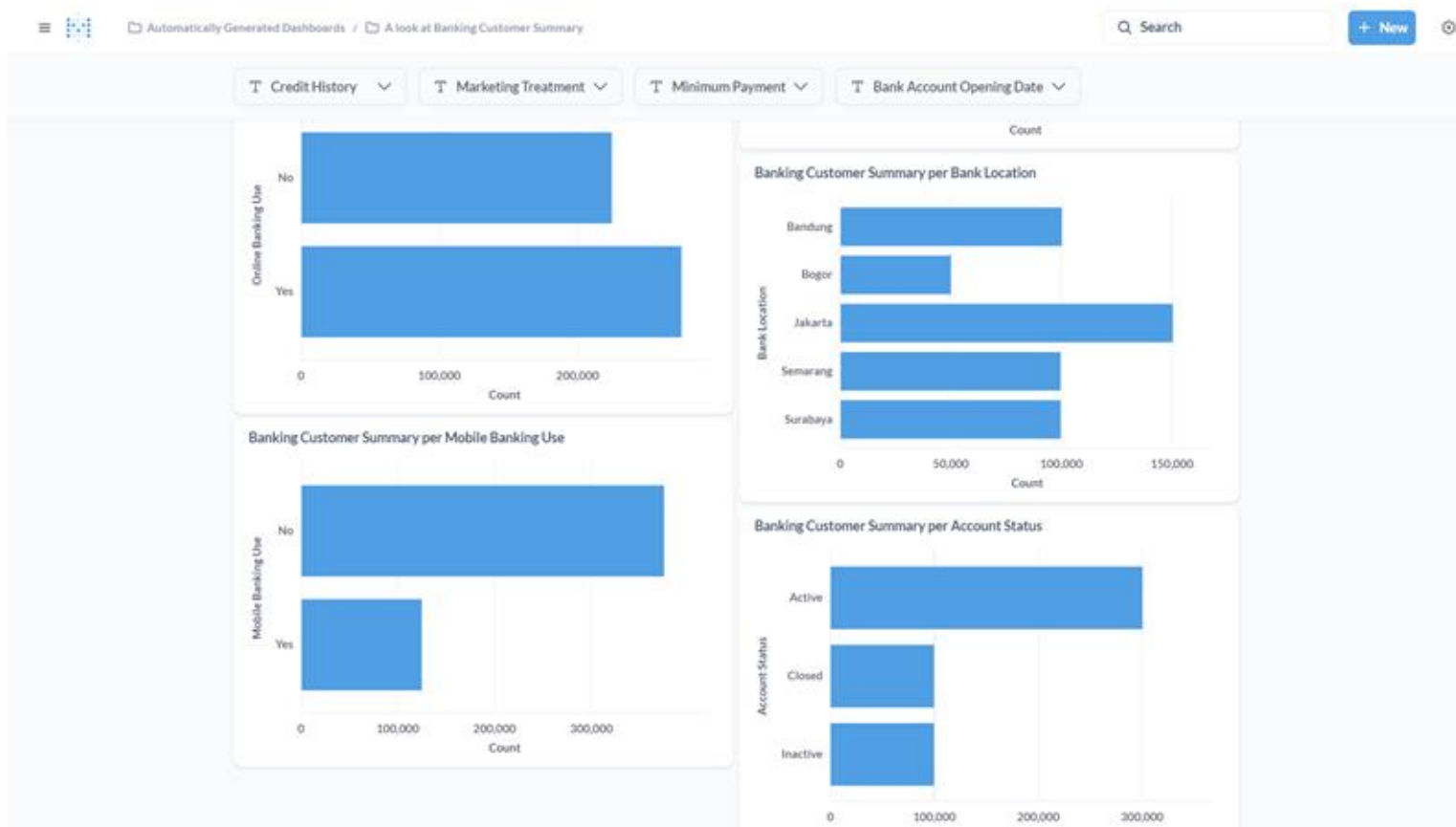
Steps to connect Metabase to a Aurora PostgreSQL database, including entering the host, port, database name, username, and password.

Metabase Dashboard Using X-Ray



Metabase has an X-ray feature that generates a summary or visualization of a table or column to provide users with a quick understanding of their data.

Metabase Dashboard Using X-Ray



A quick look at various analytics insights, such as banking customer summaries, loan terms, loan amounts, credit scores, and more.

Conclusion and Recommendation

Conclusion

- **Scalable Infrastructure:** Deploying Metabase on AWS EC2, combined with RDS Aurora PostgreSQL, provides a scalable and reliable data analytics platform.
- **Efficient Setup:** The step-by-step process ensures a smooth setup of both the database and Metabase, facilitating quick deployment.
- **Enhanced Data Visualization:** Users can leverage Metabase's capabilities for powerful data visualization and analytics.
- **AWS Integration Benefits:** The integration with AWS services ensures high availability, security, and performance for data-driven decision-making.

Recommendations

- **Security:** Use AWS Secrets Manager, enable database encryption, and manage access with security groups and IAM roles.
- **Backups and Monitoring:** Implement regular RDS backups with automated snapshots and set up CloudWatch for monitoring and alerts.
- **Cost Optimization:** Review and optimize AWS usage, selecting the right instance types and using reserved instances or savings plans.
- **Updates and Training:** Keep Metabase and AWS services updated, and ensure thorough documentation and team training.

Thank You